

3.4 Hydroxy Compounds

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.4 Hydroxy Compounds
Difficulty	Easy

Time allowed: 20

Score: /11

Percentage: /100

Question 1

Compound **X** produces a carboxylic acid when heated under reflux with acidified potassium dichromate (VI). Compound **X** has no reaction with sodium metal.

What could be the identity of compound **X**?

- A. Propanal
- B. Propanone
- C. Propan-1-ol
- D. Propan-2-ol

[1 mark]

Question 2

Considering **only** structural isomers, what is the number of alcohols of each type with the formula $C_5H_{12}O$?

	Primary	Secondary	Tertiary
A	3	3	2
B	4	2	2
C	4	3	1
D	5	2	1

[1 mark]

Question 3

Which sequence of reagents may be used in the laboratory to convert propan-1-ol into 2-bromopropane?

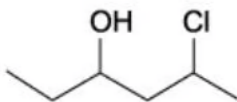
- A. Concentrated sulfuric acid, followed by bromine
- B. Concentrated sulfuric acid, followed by potassium bromide
- C. Ethanolic sodium hydroxide, followed by bromine
- D. Ethanolic sodium hydroxide, followed by potassium bromide

[1 mark]

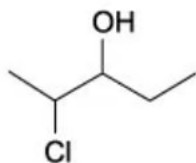
Question 4

Which diagram gives the skeletal formula of 4-chloropentan-2-ol?

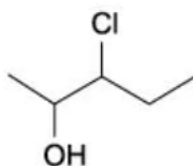
A.



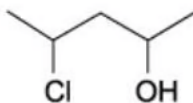
B.



C.



D.



[1 mark]

Question 5

What can be produced when an aqueous solution of butan-2-ol is oxidised under suitable conditions?

- 1 Butanone
- 2 Butanoic acid
- 3 Butanal

A. 1, 2 and 3

B. 1 and 2

C. 2 and 3

D. 1 only

[1 mark]

Question 6

Which structures show a primary alcohol that **cannot** be dehydrated to form an alkene?

- 1 CH_3OH
- 2 $\text{CH}_3\text{CH}_2\text{OH}$
- 3 $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$

A. 1, 2 and 3

B. 1 and 2

C. 2 and 3

D. 1 only

[1 mark]

Question 7

Which alcohol has a chiral centre **and** can be oxidised to a ketone?

- A. Pentan-2-ol
- B. Pentan-3-ol
- C. 3-methylhexan-1-ol
- D. 3-methylhexan-3-ol

[1 mark]

Question 8

Which reaction will distinguish between a primary and a secondary alcohol?

- A. Warming with $\text{H}^+ / \text{MnO}_4^-$
- B. Warming with $\text{H}^+ / \text{Cr}_2\text{O}_7^{2-}$
- C. Dehydration, followed by reaction with $\text{Br}_2(\text{aq})$
- D. Oxidation, followed by reaction with Fehling's (or Tollens') reagent

[1 mark]

Question 9

An alcohol with molecular formula $\text{C}_n\text{H}_{2n+1}\text{OH}$ has a chiral carbon atom but does not react with hot, acidified $\text{K}_2\text{Cr}_2\text{O}_7$.

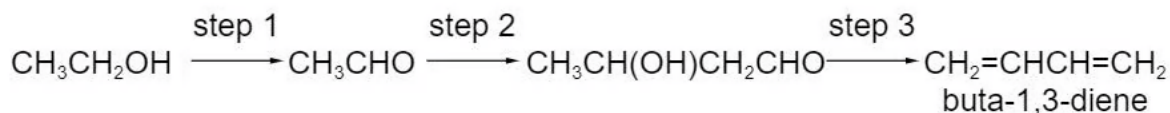
What is the smallest possible value for n ?

- A. 5
- B. 6
- C. 7
- D. 8

[1 mark]

Question 10

Buta-1,3-diene is currently obtained from fossil fuel sources. In future it may be obtained from ethanol, which can be produced from non-food agricultural crops. The sequence of reactions is as follows.



Which term could be used to describe step 1?

- A. Condensation
- B. Dehydration
- C. Dehydrogenation
- D. Hydrogenation

[1 mark]

Question 11

The functional group in a primary alcohol is $-\text{CH}_2\text{OH}$.

Which reagent reacts with a primary alcohol, under suitable conditions, to give an organic product with the same number of oxygen atoms as the alcohol?

- A. Al_2O_3
- B. $\text{CH}_3\text{CO}_2\text{H}$
- C. HBr
- D. Na

[1 mark]

Model Answers: Easy

1

The correct answer is **A** because:

- When an aldehyde is heated under reflux with acidified potassium dichromate (VI), a carboxylic acid is produced.
 - This is an oxidation reaction.
- Propanal is an aldehyde, so oxidation of this compound would produce a carboxylic acid.
- Propanal does not react with sodium metal.

B & D are incorrect as propanone is a ketone, and propan-2-ol is a secondary alcohol – secondary alcohols are oxidised to ketones, and ketones do not oxidise further.

C is incorrect as propan-1-ol is a primary alcohol. Primary alcohols are oxidised to aldehydes and then to carboxylic acids. Alcohols react with sodium metal, so, compound **X** cannot be propan-1-ol.

2

The correct answer is **C** because:

- Since the question is only asking for structural isomers, we do not need to consider stereoisomers.
- **Primary**
 - Pentan-1-ol
 - 2-methylbutan-1-ol
 - 3-methylbutan-1-ol
 - 2,2-dimethylpropan-1-ol
- **Secondary**
 - Pentan-2-ol
 - Pentan-3-ol

- 3-methylbutan-2-ol

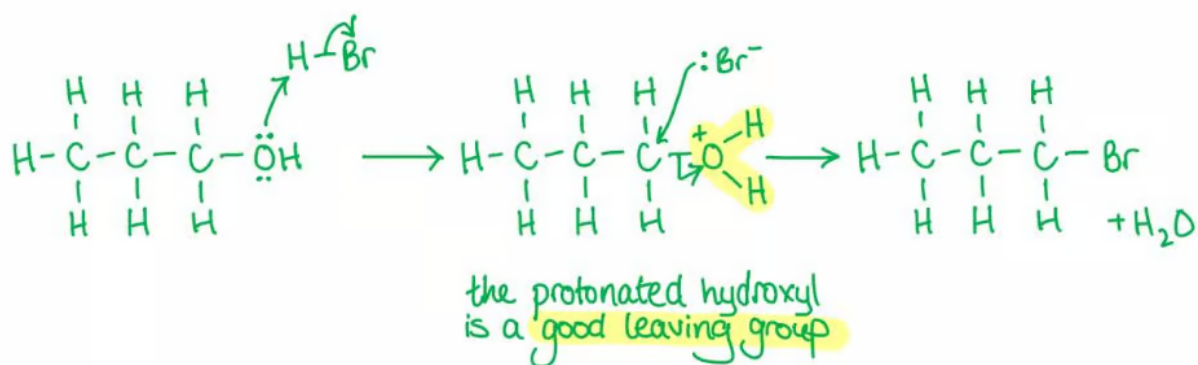
- **Tertiary**

- 2-methylbutan-2-ol

3

The correct answer is **B** because:

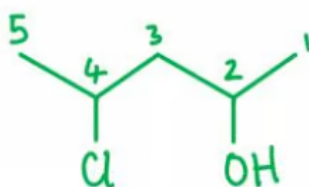
- Instead of using hydrobromic acid, propan-1-ol can be treated with a mixture of sodium or potassium bromide and concentrated sulfuric acid.
 - $\text{H}_2\text{SO}_4 + 2\text{KBr} \rightarrow \text{K}_2\text{SO}_4 + 2\text{HBr}$
- The hydrogen bromide then reacts with the alcohol.
- The reaction mixture is then warmed to distil off the bromoalkane.



4

The correct answer is **D** because:

- The best approach to questions like this is to number the carbon chains.
- Generally, the lowest possible numbers want to be used, then groups are numbered in order of priority.
- The hydroxyl (OH) group is a higher priority group and therefore has a lower number, 2.
- Chlorine therefore takes the higher number, 4.



- The compound is called 4-chloropentan-2-ol.
 - Pentane has 5 carbons, so draw and label this first.
 - Then, the -OH group will be on carbon 2, and the Cl will be on carbon 4.

A is incorrect as this compound has 6 carbons. It is 5-chlorohexan-3-ol.

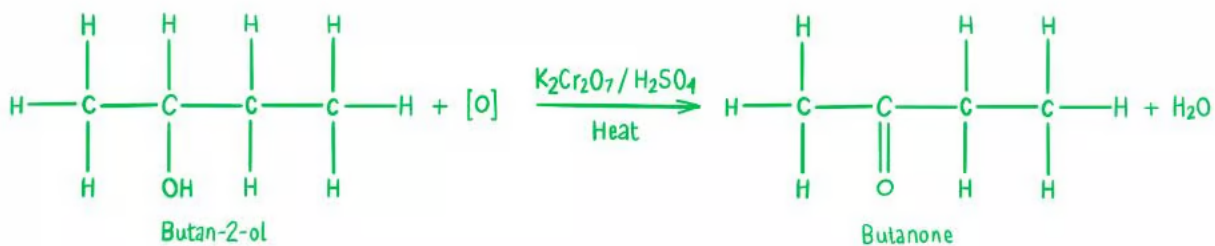
B is incorrect as this compound is 4-chloropentan-3-ol.

C is incorrect as this compound is 3-chloropentan-2-ol.

5

The correct answer is **D** because:

- Butan-2-ol is a secondary alcohol.
- It is possible to oxidise alcohols under the following conditions:
 - **Acidified potassium dichromate**, where the dichromate ion ($\text{Cr}_2\text{O}_7^{2-}$) is the oxidising agent: written as $\text{H}^+/\text{Cr}_2\text{O}_7^{2-}$ or $\text{H}_2\text{SO}_4/\text{K}_2\text{Cr}_2\text{O}_7$
 - **Acidified potassium permanganate** under **reflux**, where the permanganate ion (MnO_4^-) is the oxidising agent: written as $\text{H}^+/\text{MnO}_4^-$ or $\text{H}_2\text{SO}_4/\text{KMnO}_4$
- When a **secondary alcohol** is oxidised it produces a **ketone**.
- Ketones are **not** oxidised further.
- Therefore, only **butanone** (a ketone) is produced.



A, B & C are incorrect as aldehydes such as butanal and carboxylic acids such as

Butan-2-ol

Butanone

A, B & C are incorrect as aldehydes such as butanal and carboxylic acids such as butanoic acid can only be produced via oxidation of a **primary** alcohol, in this case butan-1-ol.

6

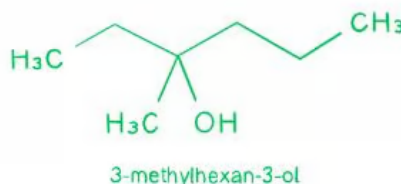
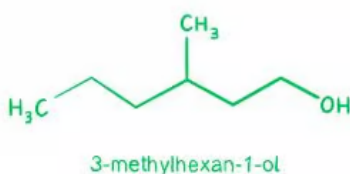
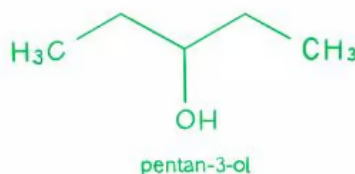
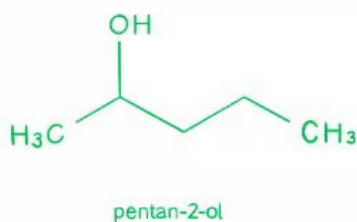
The correct answer is **D** because:

- **Dehydration** involves the **loss of water** from a single reactant.
- Alcohols undergo dehydration to form an **alkene**.
- In order to do this, a **C=C double bond** must form.
- Methanol only has one carbon, hence it cannot be dehydrated to form an alkene.

7

The correct answer is **A** because:

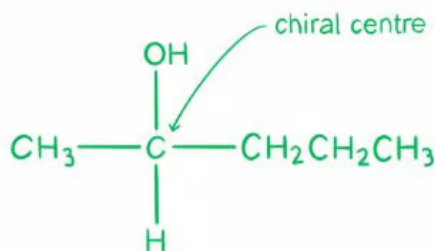
- First, let's draw the skeletal formulae of each compound.



- A ketone is produced when a **secondary alcohol** is oxidised.
 - Hence, this eliminates options **C** (a primary alcohol) and **D** (a tertiary alcohol).
- A chiral carbon atom is a carbon atom that is attached to four different types of

atoms or groups of atoms.

- The only compound with a chiral centre is pentan-2-ol:



- Therefore, **A** is correct.

B is incorrect as pentan-3-ol is a secondary alcohol, but it does not have a chiral centre.

C is incorrect as 3-methylhexan-1-ol is a primary alcohol.

D is incorrect as 3-methylhexan-3-ol is a tertiary alcohol.

8

The correct answer is **D** because:

- To distinguish between primary and secondary alcohols you would use either **Tollen's reagent** or **Fehling's solution**.
- The unknown alcohol would **first need to be oxidised** to either the ketone or the aldehyde.
 - Primary alcohols are oxidised to **aldehydes** (and then to carboxylic acids).
 - Secondary alcohols are oxidised to **ketones**.
- Tollen's or Fehling's would be added, these are both **oxidising agents** which are themselves reduced, causing a colour change.
 - Aldehydes **will** react because they are further oxidised to a carboxylic acid.
 - Ketones will **not** react because they are not oxidised.
- For Tollen's reagent, a positive result produces a **silver mirror**.
- For Fehling's solution, a positive result will cause a blue (CuSO₄ (aq)) to red (Cu₂O (s)) colour change as the copper is reduced.

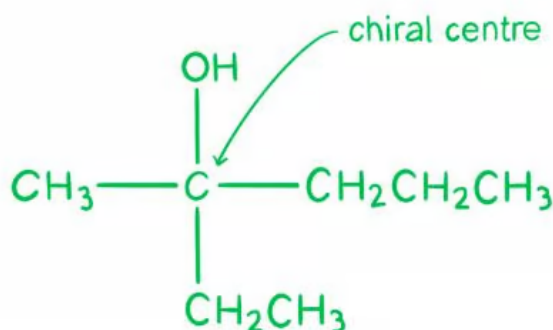
A & B are incorrect as acidified KMnO_4 and KCr_2O_7 oxidise **both** primary and secondary alcohols. Therefore, we would not be able to distinguish between the products of either without further tests.

C is incorrect as **both** primary and secondary alcohols can be dehydrated to form alkenes. The test with bromine water $\text{Br}_2(\text{aq})$ only tells us about the presence of a $\text{C}=\text{C}$ double bond, nothing else.

9

The correct answer is **C** because:

- Since tertiary alcohols do not react with dichromate (IV) ions, the alcohol must be a **tertiary alcohol**.
 - So, the basic structure of the compound is a carbon with a hydroxy group, $-\text{OH}$, and three other alkyl attached to it.
- We are told the compound has a chiral centre around C, so, each group attached must be different.
- In order to have the smallest possible value of n:
 - The first bond must be with OH
 - The second bond must be with a methyl group (1 carbon)
 - The third bond must be with an ethyl group (2 carbons)
 - The fourth bond must be with propyl group (3 carbons)
- This makes a total of 7 C atoms.



10

The correct answer is **C** because:

- Ethanol $\text{CH}_3\text{CH}_2\text{OH}$ is a **primary alcohol**.
- Ethanal CH_3CHO is an **aldehyde**.
- Aldehydes can be obtained from primary alcohols by one of two methods:
 - By oxidation with acidified potassium dichromate.
 - By **catalytic dehydrogenation** over **hot copper**.

10

The correct answer is **C** because:

- Ethanol $\text{CH}_3\text{CH}_2\text{OH}$ is a **primary alcohol**.
- Ethanal CH_3CHO is an **aldehyde**.
- Aldehydes can be obtained from primary alcohols by one of two methods:
 - By oxidation with acidified potassium dichromate.
 - By **catalytic dehydrogenation** over **hot copper**.
- Oxidation is not listed, so the answer is dehydrogenation.

A is incorrect as condensation causes a bond to break, producing 2 organic molecules and a water molecule.

B is incorrect as dehydration is the loss of a water molecule causing a double bond to be formed.

D is incorrect as hydrogenation is the addition of hydrogen to a molecule, in this example there is a loss of hydrogen.

11

The correct answer is **D** because:

- When a primary alcohol reacts with sodium, a salt (sodium alkoxide) and hydrogen gas are formed.
- Let's look at the reaction between methanol and sodium.
 - $2\text{CH}_3\text{OH} + 2\text{Na} \rightarrow 2\text{CH}_3\text{ONa} + \text{H}_2$
- It can be seen the number of oxygens does not change.
 - Therefore, the answer is **D**.

A is incorrect as $\text{CH}_3\text{CH}_2\text{OH} \xrightarrow{\text{Al}_2\text{O}_3} \text{CH}_2=\text{CH}_2 + \text{H}_2\text{O}$
1 oxygen → 0 oxygen

B is incorrect as $CH_3OH + CH_3CO_2H \longrightarrow CH_3CO_2CH_3 + H_2O$
1 oxygen $\rightarrow$ 2 oxygen

C is incorrect as $CH_3OH + HBr \longrightarrow CH_3Br + H_2O$
1 oxygen $\rightarrow$ 0 oxygen

